SiRA: Sparse Mixture of Low Rank Adaptation

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Abstract

Parameter Efficient Tuning has been an prominent approach to adapt the Large Language Model to downstream tasks. Most previous works considers adding the dense trainable parameters, where all parameters are used to adapt certain task. We found this less effective empirically using the example of LoRA that introducing more trainable parameters does not help. Motivated by this we investigate the importance of leveraging "sparse" computation and propose SiRA: sparse mixture of low rank adaption. SiRA leverages the Sparse Mixture of Expert(SMoE) to boost the performance of LoRA. Specifically it enforces the top k experts routing with a capacity limit restricting the maximum number of tokens each expert can process. We propose a novel and simple expert dropout on top of gating network to reduce the over-fitting issue. Through extensive experiments, we verify SiRA performs better than LoRA and other mixture of expert approaches across different single tasks and multitask settings.

1 Introduction

Large Language Models (Thoppilan et al., 2022; Passos et al., 2023; Ouyang et al., 2022; Touvron et al., 2023) (LLMs) have demonstrated impressive capabilities in a wide range of tasks. Yet to adapt these general-purpose models to downstream low resource tasks is especially important. To this end parameter efficient tuning (PET) (Hu et al., 2021; Li and Liang, 2021; Lester et al., 2021; Houlsby et al., 2019; Zhang et al., 2023; Zaken et al., 2021; Chen et al., 2022), which introduces task specific weights to the frozen foundation model for gradient descent, avoids catastrophic forgetting (Luo et al., 2023) of fine-tuning and offers better quality and lower cost than in-context learning (Liu et al., 2022). LoRA (Hu et al., 2021) is a widely adopted PET method which achieves high performance leveraging low-rank matrices.

One question for user of LoRA or other PET approach is how much trainable parameter should be used. In the case of LoRA, it is controlled by the rank of the low-rank matrices. And by increasing this value more computation could be provided to fit specific tasks. However it has been shown higher rank matrices will not introduce better quality to the model due to the instability of training (Chen et al., 2022), which we verify in Figure 2 in Appendix 8.1. This poses a hidden bottleneck for the quality of the model even when we have enough computation budget, and it remains a challenge to improve the quality of LoRA.

On the other hand, leveraging Sparse Mixture of Experts in neural networks has been extensively studied as a replacement for FeedForward Networks, with different approaches to find the optimal assignment between experts and tokens, including reinforcement learning (Bengio et al., 2015), linear programs (Lewis et al., 2021), fixed rules (Roller et al., 2021), top-1 gating (Fedus et al., 2022), top-2 gating (Lepikhin et al., 2020), top-k gating (Shazeer et al., 2017), reverse expert choosing (Zhou et al., 2022), and soft assignment (Puigcerver et al., 2023). Several recent works have proposed to utilize mixture-of-expert models on top of parameter-efficient tuning (Wang et al., 2022; Zadouri et al., 2023). However, prior works overlooked the potential of sparse MoE (SMoE), arguably because of issues like token dropping (Puigcerver et al., 2023) and overfitting (Elbayad et al., 2022).

To this end, we investigate leveraging "sparse" computation for PET in this paper and propose **SiRA**, the Sparse Mixture of Low Rank Adaptation. SiRA enforces the top k experts routing, which improves resource and computation utilization and also facilitates more fine-grained capacity control of a given input. SiRA consists

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of three important ingredients: the capacity constraint which embraces token dropping, an auxiliary loss to penalize over-utilizing few experts, and a novel expert dropout mechanism. They work together to ensure the proper load balancing and address the over-fitting issue.

We conducted extensive experiments to verify the performance of SiRA, which achieves better performance than LoRA (Hu et al., 2021) and its MoE variants Adamix (Wang et al., 2022), MoLoRA (Zadouri et al., 2023) across wide range of single task and multitask benchmarks. Our ablation study showed that the number of used experts and capacity per expert improves performance which demonstrates advantage of being "sparse". Notably the expert dropout plays an important role, and it is more effective than SMoE-dropout (Chen et al., 2023a).

2 Related Works

Parameter Efficient Tuning (PET) Parameter Efficient Tuning has a variety of flavors such as Adapters (Houlsby et al., 2019), Prefix Tuning (Li and Liang, 2021; Liu et al., 2021), Prompt Tuning (Lester et al., 2021), P-tuning (Liu et al., 2023), attention-injection (Zhang et al., 2023), LoRA (Hu et al., 2021; Dettmers et al., 2023), and combinations of PET methods (Mao et al., 2021). In this paper, our focus on LoRA as it has been found to achieve better results, although the methods could be applied to other flavors as well.

MoE for PET methods Along the intersection of PET and MoE, Adamix (Wang et al., 2022) and MoLoRA (Zadouri et al., 2023) are most similar to our work. Adamix randomly chooses an expert in training and averages all the experts during inference. Albeit efficient, this method is more like checkpoint averaging (Gao et al., 2022) because the experts are randomly chosen they don't learn to specialize. More importantly, the random approach has significantly longer training time, which is multiplied by the number of experts used. MoLoRA applies the full soft MoE on the top of LoRA, where all experts are averaged using a learned gating. Compared to this work, our method can achieve better efficiency. Firstly, SiRA does not need longer time to train compared to standard LoRA, thanks to the quick convergence of SMoE (Fedus et al., 2022). Secondly, the sparsity is enforced in SiRA which saves the training resource and inference computation compared to MoLoRA.

Another track of the MoE work is for multitasking, such as Task-MoE (Kudugunta et al., 2021) and Skill Selection (Ponti et al., 2023). These approaches assume the external task-id as an extra input for training and inference. Although we experiment with MoE in multitask settings, it does not require the task-id of inputs.

3 Sparse Mixture of Low Rank Adaptation

To increase the capacity of LoRA (Hu et al., 2021) using Mixture of Experts (MoE) without adding too much computational cost, we propose Sparse Mixture of Experts of Low Rank Adaptation (SiRA), which leverages multiple lightweight LoRA adaptors as experts while enforcing sparsity when using the experts.

Figure 1 shows an illustration of SiRA. The MoE layer for the adapter consists of E experts, each with their own LoRA weights, $W_1, ..., W_E$. W_k is the product of two low rank matrices $W_k = B_k A_k$. We also assume the base foundation model has W_0 as it is frozen weight, which represents either query, key, value, or output projection. Next, we will introduce each component of our framework that enforces the sparsity of LoRA experts.

Expert Gating To reduce the computational cost, SiRA only actives a subset of all the expert modules. Formally, during each forward pass, we select K out of E experts using the output scores of a gating network θ_g. The process is mathematically expressed as Equation (1) and (2), where s denote the token index of the sequence x and $G_{s,e}$ is the gating network output at s-th token e-th experts.

$$G(x_s) = \text{TopK}(\text{softmax}(\theta_g^T x_s)) \tag{1}$$

$$y_s = \sum_{e=1}^{E} G_{s,e} W_e(x_s) + W_0(x_s) \tag{2}$$

Experts Dropout We propose a practical way to make the gate more balanced with a simple gate dropout. Specifically, we introduce dropout to the gating output G as shown in Equation 3.

$$G(x_s) = \text{TopK}(\text{Dropout}(\text{softmax}(\theta_g^T x_s))) \tag{3}$$

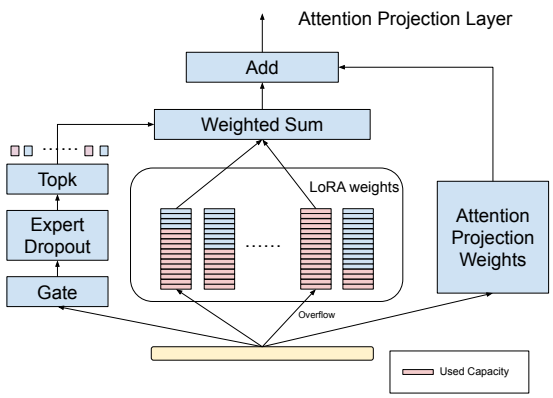

Figure 1: SiRA: Sparse Gated Mixture of LoRA.

Expert Token Capacity We enforce the capacity constraints for experts following GShard (Lepikhin et al., 2020). Specifically, we restrict that the number of tokens processed by each expert should not exceed a predefined threshold. Once the capacity is reached, one expert simply drop the overflow tokens. If all K experts reach their token capacity before all tokens in a training example is processed, the rest of the tokens will only be encoded using the frozen model parameter W_0.

Auxiliary Loss We define an auxiliary loss term to further encourage load balancing among different experts (Shazeer et al., 2017). We define the total number of tokens to be S, and there is E experts. We denote c_e as the number of tokens routed to expert e. By using the mean gates per expert $m_e = Mean_s(\text{Dropout}(\text{softmax}(\theta_g^T x_s)))$ as a differentiable approximation, we define the aux loss in Equation 4.

$$l_{aux} = \frac{1}{E} \sum_{e=1}^{E} \frac{c_e}{S} * m_e \tag{4}$$

4 Experiments

4.1 Evaluation Setup

Baselines and Experiment Configs We specifically compare our model with the standard LoRA (Hu et al., 2021), Adamix (Wang et al., 2022) and MoLoRA (Zadouri et al., 2023). Note that other adapter approaches are not commpared with as the SiRA approach is orthogonal and could be applied on top of them as well. We choose the PALM2-FLAN XXS (Passos et al., 2023) as the foundation model. We follow the default configurations in (Hu et al., 2021) to inject LoRA weights on the attention projections and set the intrinsic rank as 4. We use 16 experts by default across all baselines. For training config and model selection, see Appendix 8.2.

Datasets and Metrics We evaluate on the following datasets:[1]

XTREME-UP The XTREME-UP dataset (Ruder et al., 2023) is a multilingual multitask dataset, with a focus on the scarce-data scenarios of underrepresented languages. In this work, we choose two of the underrepresented languages—Swahili(SW) and Bengali(BN)—and evaluated on several NLP tasks where these two languages have training and evaluation data. We follow Ruder et al. (2023) for each task's splits and evaluation metrics.

FinQA FinQA (Chen et al., 2021) is a QA dataset in the financial domain. Complex reasoning capabilities are needed to correctly answer these questions. Note that the answers of the FinQA dataset are programs of a special arithmetic DSL. In this work we only evaluate on metrics based on surface form matching, *i.e.*, exact match and F1 scores.

ForumSum ForumSum (Khalman et al., 2021) is a diverse and high quality conversation summarization dataset with human written summaries where the conversations are collected from a wide variety of internet forums. We report BLEURT (Sellam et al., 2020), ROUGEL, and F1 scores.

4.2 Performance of SiRA

We evaluate all the single tasks performance in Table 1. Note that as FinQA is a hard task with financial reasoning, thus the exact match and f1 score is relatively low. We can notice that SiRA is outperforming all other baselines at most of the tasks, with less than 1% extra parameters. Notably when compared to MoLoRA, SiRA achieves constantly better performance among all the tasks. This demonstrates that "sparse" MoE is better than "full" MoE. Adamix shows some small advantage on the Semantic Parsing task, but overall loses to SiRA across all other tasks.

We also conducted experiments on two multitask settings on language swahili (SW) and bengali(BN), and two multiligual settings for QA in languages task and QA across languages task. We

[1]Since our base model (Chung et al., 2022) had been exposed to many public datasets during training, we choose dataset that are not consumed yet.

Table 1: Performance Comparison For Single Tasks

Approach	δ Params	FinQA (EN)		ForumSum (EN)			SP (SW)	QA-in (SW)	NER (SW)	SP (BN)	QA-in (BN)	QA-cross (BN)
		em	f1	bleurt	rougeL	f1	accuracy	f1	span-f1	accuracy	f1	f1
LoRA	0.043%	5.0	5.6	96.70	33.97	23.54	27.63	82.08	88.95	33.52	80.34	76.81
Adamix	0.689%	5.6	6.0	95.95	35.10	23.88	**33.22**	81.24	89.00	**39.03**	81.70	76.07
MoLoRA	0.746%	5.6	6.4	97.05	34.37	24.79	32.50	82.33	89.33	36.28	79.06	76.75
SiRA	0.746%	**5.8**	**6.6**	**97.14**	**35.67**	**25.83**	32.52	**83.00**	**89.95**	38.61	**82.10**	**76.93**

Table 2: Performance Comparison For Multi Tasks

Approach	δ params	SW Multitask				BN Multitask			
		SP(accuracy)	QA-in(f1)	NER(span-f1)	Average	SP(accuracy)	QA-in(f1)	QA-cross(f1)	Average
LoRA	0.043%	28.06	77.71	88.28	64.69	32.06	79.27	75.03	62.12
Adamix	0.689%	**35.14**	76.99	89.01	67.10	**38.41**	79.49	75.09	64.33
MoLoRA	0.746%	33.44	79.91	88.92	65.66	35.98	78.14	76.37	63.49
SiRA	0.746%	33.98	**81.26**	**89.04**	**68.10**	37.71	**82.17**	**75.50**	**65.13**

Table 3: Performance Comparison For Multilingual Tasks.

Approach	δ params	QA-in (9)	QA-cross (25)
LoRA	0.043%	85.09	69.41
Adamix	0.689%	84.75	70.42
MoLoRA	0.746%	85.14(WIP)	69.70(WIP)
SiRA	0.746%	**86.38**	**70.86**

Table 4: Self ablations on the hyper-parameter topK(K) and expert capacity(C) on ForumSum.

Configs	bleurt	rougeL	f1
K=2	96.87	34.51	24.73
K=4	96.60	34.66	25.34
K=6	96.75	34.73	24.55
K=8	96.76	**35.31**	**25.64**
K=10	**97.51**	35.10	25.19
K=12	96.96	34.49	24.24
C=2	96.33	34.15	24.13
C=4	96.60	34.66	25.34
C=6	97.14	**35.67**	**25.83**
C=8	**97.31**	34.97	25.24
C=10	97.25	34.75	25.57
C=12	96.50	34.44	23.94

reported numbers in Table 2 and Table 3 respectively. The overall trend is similar to what we found in the single tasks. SiRA achieves the best average performance among all baselines.

4.3 Ablation Study

Computation Ablations We share the ablations on ForumSum in Table 4. We choose a simple config as base (k=4, C=4) and then change each of them while keeping the rest. We first range the top K from 2 to 12, with capacity $C = K$. And then we fix the $K = 4$, and range the expert capacity from 2 to 12. An interesting finding is increasing the computation or the capacity will not always increase the scores and there is a "comfortable zone" which we need to find out with model tuning. This also justifies why the "full" MoE based approach is not as good as SiRA. SiRA provides more fine-grained control on the computation.

Table 5: Gating ablations on ForumSum.

Approach	bleurt	rougeL	f1
SiRA	**97.14**	**35.67**	**25.83**
- aux loss	96.37	35.09	25.11
- Expert Dropout	97.09	34.73	24.55
+ SMoE-Dropout	96.30	34.24	24.32

Gating ablations We also provide the ablations on the gating in Table 5. Specifically we compare SiRA with 3 cases: 1/ removing the aux loss, 2/ removing the gate dropout, and 3/ using a static routing based dropout SMoE-Dropout (Chen et al., 2023a) instead. Results suggested that the learned gating is still better than a fixed one, and both the gate dropout and aux loss help the performance of SiRA.

What the gate learns We use the Swahili multitask experiment to study what the gate is learning. We measure the average entropy of each gate weight distribution before TopK is applied. The average entropy for the QA (in language) task decreases from 1.6 to 1.13 nats during training. This indicates that the model learns to give certain gates more weight as it trains.

We also measure the average correlation coefficients between each task index and each gate index similar to (Chen et al., 2023b). We convert the task index to a one hot encoding for this. At the end of training, the average correlation was about

.025, which is not significant. The correlation between gates and languages in the multilingual experiment is not significant either. This suggests that our gating mechanism does not learn to route different tasks to different gates.

5 To-Do List

This manuscript is currently under active development. Our upcoming endeavors include getting more results and analysis, and improving the writings. We warmly welcome suggestions, insights, and constructive criticism from the research community. Should you have any feedback or ideas that could potentially enhance the quality and impact of our work, please do not hesitate to contact the lead author. Your input is invaluable to us, and we are eager to integrate diverse perspectives to refine and advance our research.

6 Conclusion

This paper introduced SiRA, a Sparse Mixture of Expert variant of LoRA. SiRA enforces the top k experts routing with capacity constraint for each experts. We also devise a novel expert dropout mechanism on top of the auxiliary loss to reduce its over-fitting issue. We conducted extensive experiments to verify the performance of SiRA, which achieves better performance than the LoRA and its MoE variants across different single tasks and multitask settings.

7 Limitation

SiRA is taking extra serving overhead for serving with extra parameters on experts and the gating, compared to LoRA or Adamix. How to minimize the serving overhead is a challenge problem which we hope to address our future works.

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8 Appendix

8.1 Effect of LoRA rank

We investigate the effect of LoRA rank in Figure 2.

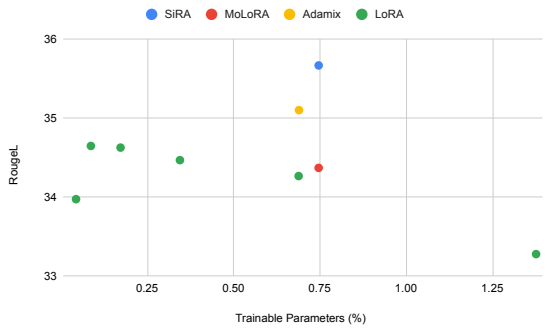

Figure 2: SiRA vs LoRA on ForumSum Task. We increase the rank of LoRA (rank=4, 8, 16, 32, 64, 128) and report the RougeL as a metrics. Notably increasing the rank does does not help the performance. SiRA (rank=4) can achieve higher quality by leveraging the sparse mixture of experts.

8.2 Training and Model selection

During supervised finetuning, SFT, we use 8 Tensor Processing Units (TPU) V3 chips for fine-tuning. The batch size is 64, and the maximum training step is 30000. We use the Adafactor optimizer (Shazeer and Stern, 2018) with a learning rate of 0.0005. Both the input and output sequence lengths are set to match the dataset requirements. The training dropout rate is 0.05. The expert dropout rate is set to 0.5. We did hyper-parameters search to find the best model configurations. We decode on the validation sets of each task every 100 steps. And we report test results from the best checkpoints according to the validation scores. For multitask results, the checkpoint is picked on the average each tasks metrics. For the reported numbers in section 4.2, we use topk $K = 4$ as default. Yet we found $K = 8$ is better for BN multitask and QA (in-lang) multilingual setting, and $K = 12$ better for QA (cross-lang) experiments.